NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED **MATHEMATICS** **M.M. 63**

REVISION ASSINGMENT # 09 (DETERMINANT AND MATRIX) TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. A square matrix P satisfies $P^2 = I - P$ where I is an identity matrix. If $P^n = 5I - 8P$, then n is-

(A) 4 (B) 5 (C) 6 (D) 7

2. If maximum and minimum values of the determinant $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$ are α and β respectively then which one of the option is incorrect ?

(A) $\alpha + \beta^{99} = 4$

(B) $\alpha^3 - \beta^{17} = 26$

(C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in N$

(D) A triangle can be constructed having it's sides as α, β and $5\alpha - 13\beta$.

3. If a_0, a_1, a_2, a_3, a_4 are in A.P. with commom difference d (where $d \neq 0$ & ± 1) , then the value of

$\begin{vmatrix} a_1a_2 & a_1 & a_0 \\ a_2a_3 & a_2 & a_1 \\ a_3a_4 & a_3 & a_2 \end{vmatrix}$ is

(A) $2d^4$ (B) $2d^3$ (C) $2d^2$ (D) $4d^4$

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4. Which of the following statement is correct about two square matrices A and B of same order -

(A) trace (Adj(AB)) = Adj(trace (AB).I)

(B) trace ((A + B) (A – B)) ≠ trace (A^2) – trace(B^2)

(C) trace(Adj(|A| |B| AB)) – trace(Adj(|AB| BA)) = 0

(D) If A is symmetric matrix and B is skew symmetric matrix then trace (AB' – BA') ≠ 0

Space for Rough Work

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5. If A is a skew symmetric matrix of order 2 and B, C are $\begin{bmatrix}1 & 4\\2 & 9\end{bmatrix}, \begin{bmatrix}9 & -4\\-2 & 1\end{bmatrix}$ respectively, then $A(B^2C^2)A^T$ is -

(A) a symmetric matrix

(B) an identity matrix

(C) a skew symmetric matrix

(D) neither symmetric nor skew symmetric

6. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $|A + B|$ is -

(A) 0 (B) –1 (C) 1 (D) 2

7. If A is a square matrix of order 'n' with $|A| = 9$ and $|adjA| = 3^{16}$, then number of elements in A, is -

(A) 81 (B) 27 (C) 9 (D) 3

8. Number of solutions of equation $\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \cos\left(\theta+\frac{\pi}{6}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & -\sin\left(2\theta+\frac{\pi}{3}\right) \\ -\cos\left(\theta-\frac{\pi}{6}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{2\pi}{3}\right) \end{vmatrix} = 0$ in $[0,2\pi]$ is-

(A) 2 (B) 4 (C) 8 (D) infinite

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Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

9. System of equations $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0, b \in R$

(B) no solution when $a = 0, b = 1$

(C) infinite solutions when $a = 0, b = 2$

(D) infinite solutions when $a = 1, b \in R$

10. If $A = \begin{bmatrix}1 & 1 & 0\\2 & 2 & 1\\0 & 1 & -1\end{bmatrix}$ and $A^3 - 2A^2 + \lambda A + I = \mathbf{O}$ (Null matrix), then-

(A) $\lambda = 4$ (B) $\lambda = -4$ (C) $|A^2 + A^{-1}| = 56$ (D) $|A^2 + A^{-1}| = 14$

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Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 11 to 12

If A is invertible matrix of order 3 and B is another matrix of same order as of A such that $|B| = 2$ and $A^T|A|B = A|B|B^T$, then

(where adj(A) and A^{-1} denote adjoint and inverse of matrix A)

11. $\left|AB^{-1}\text{adj}\left(A^TB\right)^{-1}\right|$ is equal to-

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{16}$ (D) 1

12. If B is symmetric matrix , then which of the following is incorrect ?

(A) adj(B) is symmetric matrix (B) B^{-1} is symmetric matrix

(C) $A^{2015}\,B^{2016}$ is symmetric matrix (D) AB^{-1} is symmetric matrix if AB = BA

Paragraph for Questions 13 to 15

$$g(y) = \begin{vmatrix} p-y & q & r \\ r & p-y & q \\ q & r & p-y \end{vmatrix}, \quad p > q > r > 0$$

On the basis of above information answer the following :

13. Absolute term in g(y) can be -

(A) 2 (B) –1 (C) 0 (D) all of the above

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14. Coefficient of y in g(y) will be -

(A) independent of only p (B) independent of q as well as r

(C) independent of only r (D) none of these

15. If y_1, y_2 & y_3 are the roots of $y^3 = 1$ such that $f(y) = p + qy + ry^2$, then g(0) is equal to -

(A) $f(y_1) \cdot f(y_2) \cdot f(y_3)$ (B) $f(y_1) + f(y_2) + f(y_3)$

(C) $\frac{f(y_1)+f(y_2)+f(y_3)}{3}$ (D) $\{f(y_1).f(y_2).f(y_3)\}^{1/3}$

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SECTION–III

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. If system of equations

$x + (\sin\alpha)y + (\sin^2\alpha)z = 0$,

$x + (\cos\alpha)y + (\cos^2\alpha)z = 0$

$x + (\sin2\alpha)y + (\sin^2 2\alpha)z = 0$

has non trivial solutions, then number of distinct values of α (where $\alpha \in [0,\pi]$), is

2. Let $a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = \begin{vmatrix} x^2+x & (x-2) & 1 \\ (2x+3) & x & 3 \\ (x-5) & -1 & (x+1) \end{vmatrix}$ is an identity w.r.t x, then the value of $(a_4 + a_3 + a_2 + a_1 + a_0) - 25$ is

3. Let $A_1 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \neq 0$ & $A_2 = \begin{vmatrix} a+2b & b+3c & c+4a \\ b+2c & c+3a & a+4b \\ c+2a & a+3b & b+4c \end{vmatrix}$, then $\frac{A_2}{A_1}$ is -

4. The number of all possible symmetric matrices of order 3×3 with entries $-1,0,1$ and whose trace equals 1, is

Space for Rough Work

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REVISION ASSINGMENT # 08 (CONIC SECTION-02) MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	A	D	A	A,B,C,D	B,D	B,C,D	B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	B,D	C,D	A	C	D	B	A	D		
SECTION-II	Q.	1	2	3	4						
	A.	4.50	1.33 or 1.34	4.00	2.00						

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